Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	08 March 2026
Team ID	
Project Name	Ucab – Cab Booking System
Maximum Marks	4 Marks

Technical Architecture:

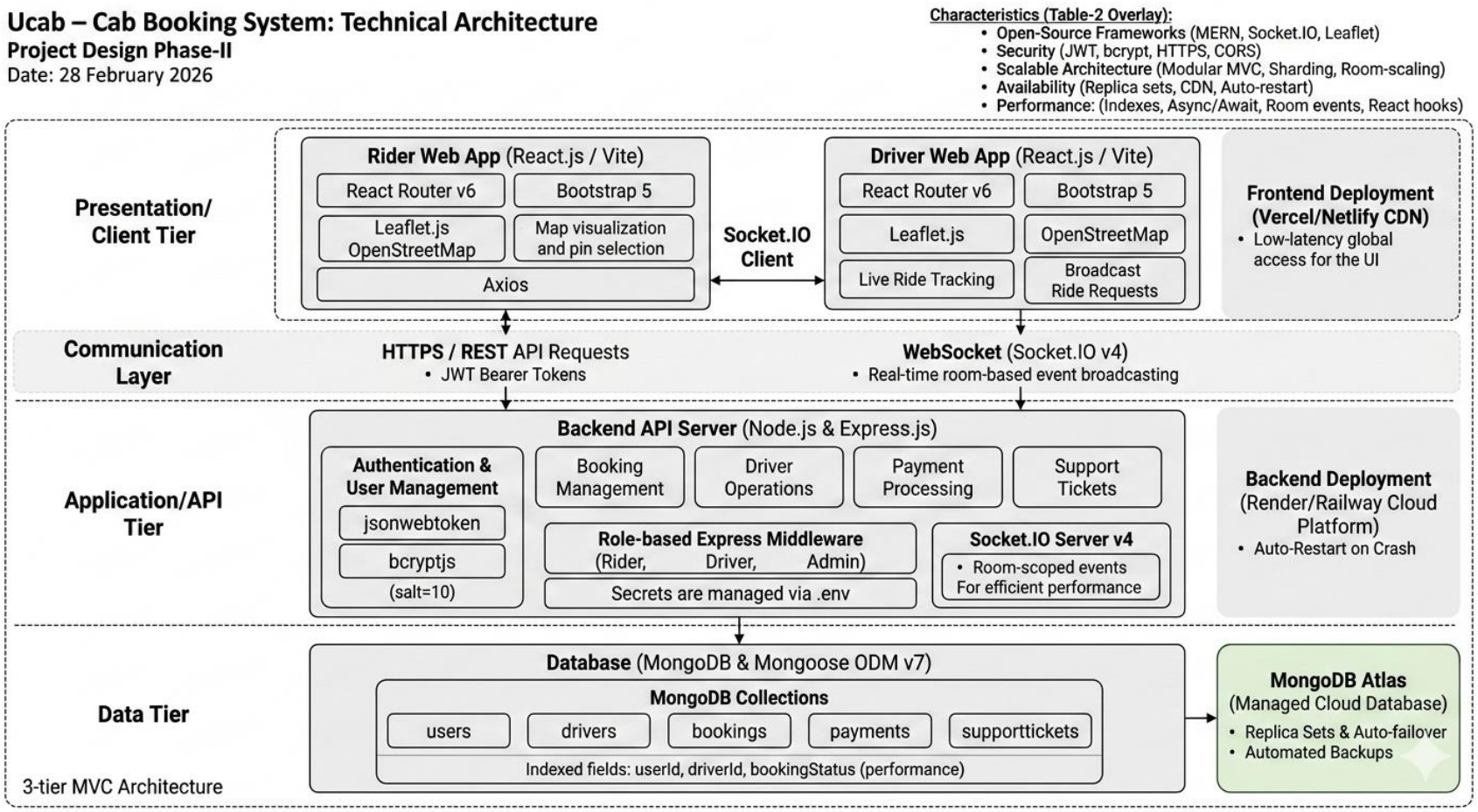

Table-1 : Components & Technologies:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

S.No	Component	Description	Technology
1.	**User Interface**	Web UI for Riders (booking, tracking, payment, history) and Drivers (dashboard, ride management). Fully responsive.	React.js (Vite), React Router v6, Bootstrap 5, Leaflet.js, Axios, Socket.IO Client
2.	**Backend API Server**	RESTful API handling auth, bookings, driver operations, payments, and support tickets. JWT-protected role-based routes.	Node.js v18+, Express.js, bcryptjs, jsonwebtoken, Mongoose ODM
3.	**Real-Time Communication**	Broadcasts ride requests to driver rooms; notifies rider on acceptance; cancels pending requests for other drivers.	Socket.IO v4 (WebSocket) — room-based event broadcasting
4.	**Database**	NoSQL document store. Collections: users, drivers, bookings, payments, supporttickets. Indexed on userId, driverId, status.	MongoDB (NoSQL) with Mongoose ODM v7
5.	**Cloud Database**	Managed cloud MongoDB with automated backups, replica sets, and auto-failover for high availability.	MongoDB Atlas — connected via MONGO_URI environment variable
6.	**Map & Location Service**	Interactive map for pickup/drop-off pin selection, nearby cab display, and live ride path rendering in browser.	Leaflet.js + OpenStreetMap (open-source, no API key required)
7.	**Infrastructure / Deployment**	Dev: Node.js (port 5000) + Vite dev server (port 5173). Production: cloud-hosted backend and CDN frontend.	Render/Railway (backend), Vercel/Netlify (frontend), .env for secrets

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	**Open-Source Frameworks**	All libraries are open-source: MERN stack, Socket.IO, Leaflet, Bootstrap, Axios, bcryptjs, jsonwebtoken.	React (MIT), Node.js (MIT), Express (MIT), MongoDB/Mongoose (Apache 2.0), Socket.IO (MIT), Leaflet (BSD-2), Bootstrap (MIT)

2.	**Security**	Passwords hashed with bcrypt (salt=10). All API routes protected by JWT Bearer tokens. Role-based Express middleware enforces Rider/Driver/Admin separation. Secrets in .env only.	bcryptjs, jsonwebtoken, role-based Express middleware, CORS, HTTPS (production)
3.	**Scalable Architecture**	3-tier MVC: React frontend / Express REST API / MongoDB. Modular route-controller pattern — each module (auth, bookings, drivers, payments, support) independently deployable.	Modular Express routes, MongoDB Atlas sharding, Socket.IO room-based scaling
4.	**Availability**	MongoDB Atlas replica sets provide automatic failover. Frontend on CDN ensures low-latency global access. Backend platform auto-restarts on crash with zero manual intervention.	MongoDB Atlas replica sets, Vercel/Netlify CDN, Render/Railway auto-restart
5.	**Performance**	MongoDB queries indexed on userId, driverId, bookingStatus for fast lookups. Axios async/await prevents UI blocking. Socket.IO broadcasts only to relevant rooms. React useState minimises re-renders.	MongoDB indexes, Axios async/await, Socket.IO room-scoped events, React useState